

Forecasting mobile network traffic report

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May 4, 2026

Abstract

This report has been written in the context of the PhD application process to join Networks Data Science group at IMDEA Networks Institute. It summarizes the answers to the tasks given to the applicant concerning time series analysis and forecasting of mobile network traffic activity in the city of Milan. The report answers the given tasks by providing text, plots, discussion, analyzes and references to support the results.

1 General methodology

For these tasks, I had any notions of time series. I had to read articles to understand how it works and the different algorithms and techniques used to forecast a time series. However, I am very familiar with Panda data analysis in general. Therefore, the first task has been relatively straightforward. **LLM usage** : LLM has been used **only** to reorganize my **resources**. I gave to chatGPT, all the websites used in this project to reorganize them in latex usable format (eg. see section Resources). However, **LMM** has not been used in any other task, neither for code generation nor for report writing.

2 Task I : characterization of the data

2.1 hardware and software setup

A brief description of the hardware and software setup used to process the data;

System	Linux system (school server dedicated for M2 students Deep Learning project accessible via ssh)
kernel version	#89-Ubuntu SMP PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025
Kernel release	6.8.0
hardware architecture	64-bit x86 processor
RAM	503 GB physical RAM and 7 GB swap space
GPU specifications	NVIDIA-SMI 470.256.02 Driver Version: 470.256.02 CUDA Version: 11.4 (Tesla K80)

2.2 Density function

A plot of the probability density function of the traffic in the target city in the target city, computed over 10,000 samples that each represent the total two-month traffic in one geographical area, and with an accompanying text where the applicant comments on the result (e.g., discussing the homogeneity or diversity of the distribution, and providing possible explanations for the observed behaviors);

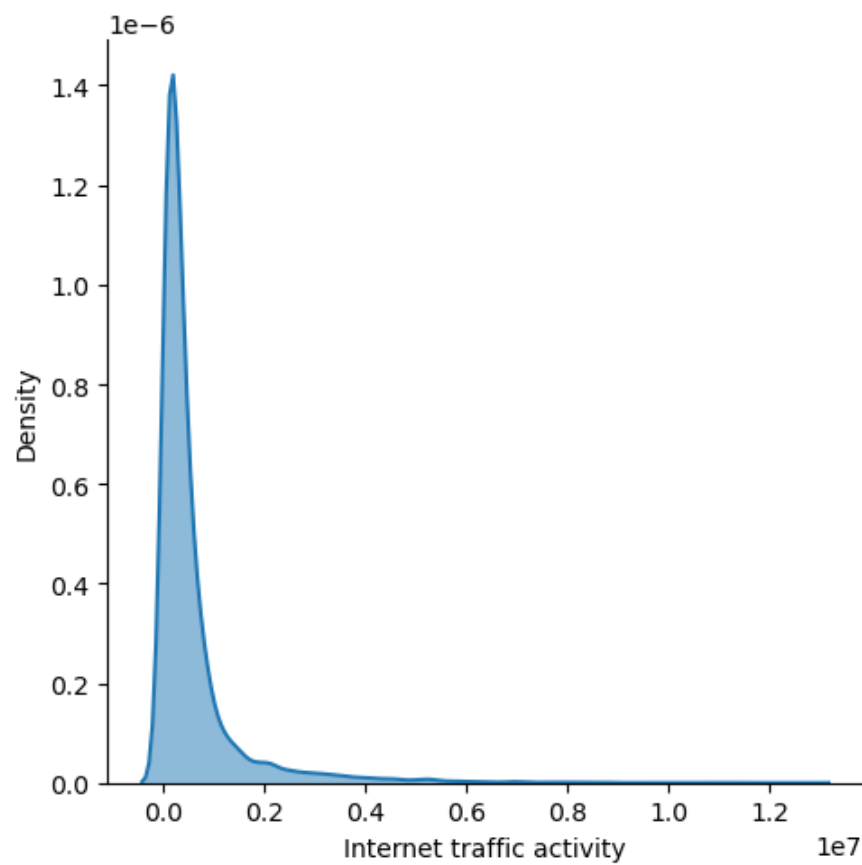


Figure 1: Internet traffic activity density over two months in one geohraphical area

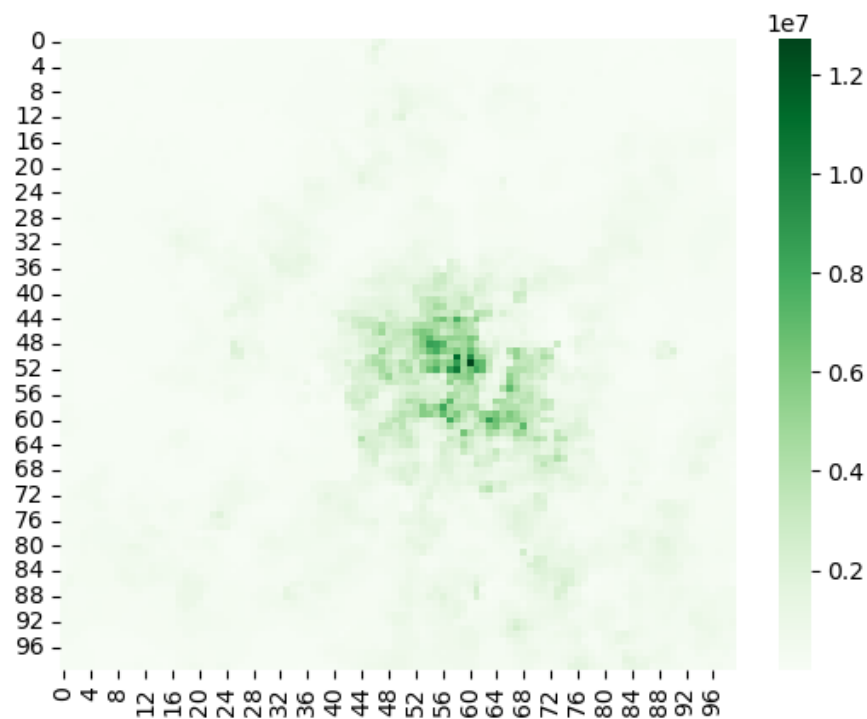


Figure 2: Heatmap two months internet traffic per area

Discussion

1. Fig. 1 shows the PDF (Probability Density Function) for the total traffic recorded in the whole geographical area for a period of two months in the city of Milan. Here, we can see that the PDF is heavy-tailed distributed. Probably, this is due to the network and data usage at different heterogeneous locations in the entire city of Milan.
2. The heat map (Fig. 2) clarifies our previous discussion, the high use of internet traffic comes from some specific areas in the center with high popularity and dense industrial and tourism activities. The use of the Internet is very heterogeneous in the city of Milan in general.

2.3 Time series figures

A figure of the time series of network traffic during the first two weeks at three areas first two weeks at three areas, namely (i) the area with the highest total traffic during the two-month period, (ii) the area with Square id 4159, and (iii) the area with Square id 4556, with an accompanying text where the applicant comments on the result (e.g., discussing the similarities or differences observed in the temporal dynamics of each area, and speculating on their causes).

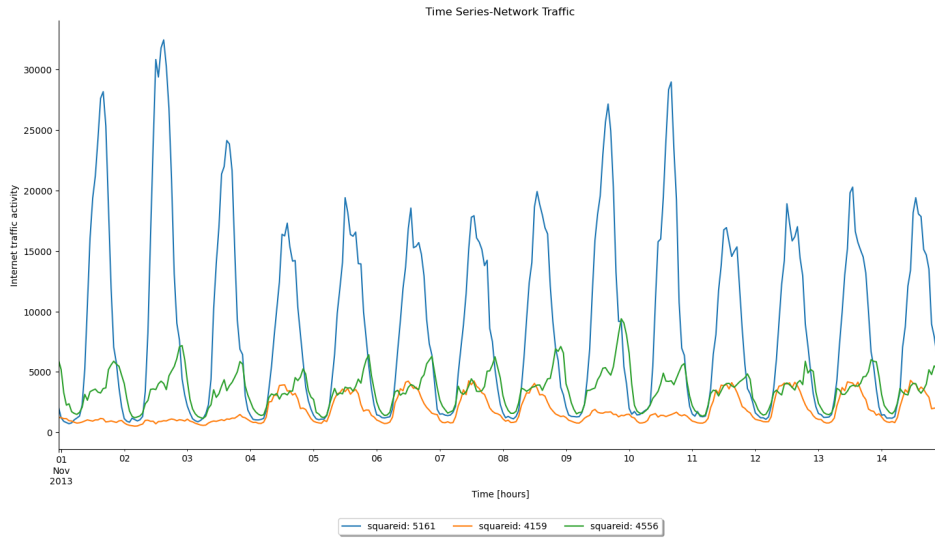


Figure 3: Network traffic during the first two weeks (hourly basis)

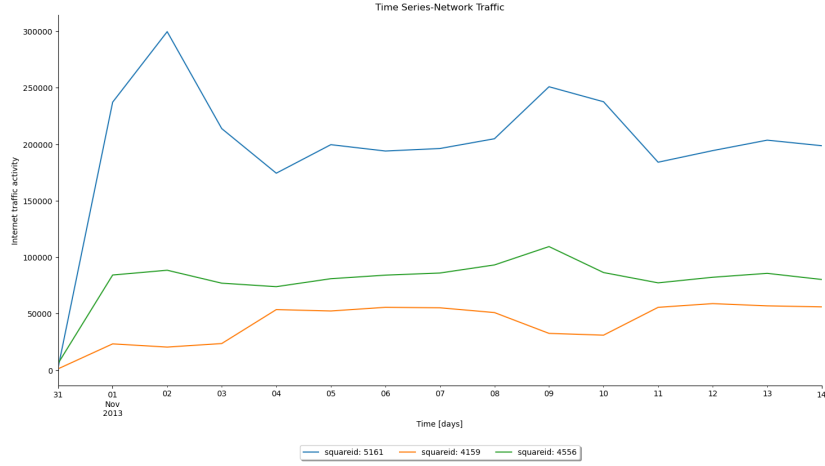


Figure 4: Network traffic during the first two weeks (daily basis)

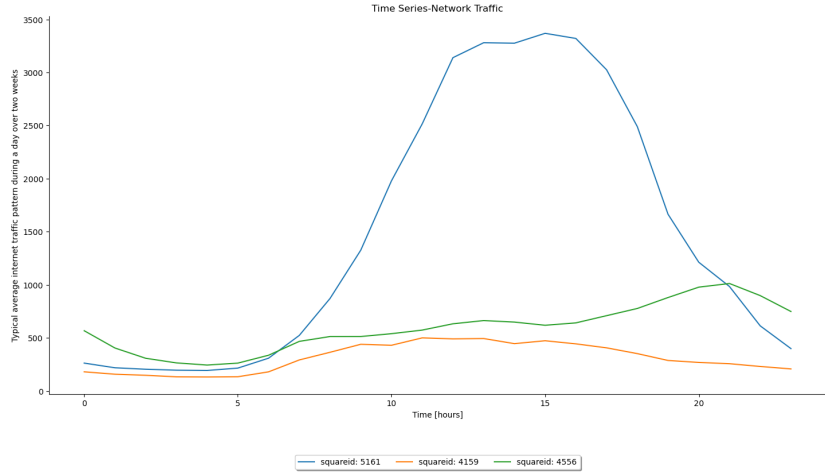


Figure 5: Average internet activity in a day for the first two weeks

Comments and observations :

1. General comments

- (a) For the three geographical areas, with the hourly basis time series, we observe a weekly seasonality for all of them, though some areas have multiple seasonality
- (b) Square id 5161 counts the highest traffic recorded for the first two weeks (an expected result)
- (c) The daily trend of internet traffic usage in the three geographical areas is quite different

2. Square id 5161

- (a) We identify daily and weekly routines of network traffic usage, there are daily and weekly seasonality

- (b) The daily usage is overall stable in the Weekdays (eg. high internet usage in the week-end and relatively stable in the Weekdays)
- (c) This area would correspond to a famous area which receives more visitors, tourists in the week-end with a relatively high commercial or industrial activity in the Weekdays

3. Square **id 4556**

- (a) This geographical area has less network traffic compared to the center cities. But, there is a slight increase in the internet traffic in the night compared to square id 5161, the daily pattern is very stable (week-end and weekdays)
- (b) This area is expected to be a lively residential neighborhood with a vibrant nightlife. People come from other cities to enjoy the nightclubs and other nighttime activities.

4. Square **id 4159**

- (a) This geographical area experiences very low network traffic in general, slow internet traffic usage around the week-end and very stable trend in the rest of weekdays
- (b) It is expected to be a school or an industrial zone for work, stable activity in the weekdays and no activity in week-end

3 Task II : one-step prediction of future traffic in a single area

3.1 Description of the proposed algorithm

A self-contained description of the proposed algorithm;

1. **Hint** : LSTM is a Deep Learning Forecasting technic which deals well with long seasonal and non-stationnary time series. We will use it for one step prediction first and re-use it recursively for multi steps prediction. For the size of our dataset, **hourly** prediction is interesting
2. **Goal** : predict one step (next hour) Internet traffic activity with a 24 window size
3. **Input**

- (a) x_t [24 size vector; numpy array] $\leftarrow$ past 24 hours Internet traffic activity history
- (b) a [int] $\leftarrow$ square id of the concerned geographical area
- 4. def LSTM(a): # train the model for area a
 - (a) Get trainable data for area a
 - (b) Pre process it by scaling (Min max scaler)
 - (c) Train the LSTM model
 - (d) return trained LSTM model (and print it training statistic)
- 5. def $f(x_t, a)$:
 - (a) return LSTM(a).predict(x_t)
- 6. **Output**
 - (a) $\hat{x}_a(t+1) = f(x_t, a)$

3.2 Plots

Three plots reporting the superposed time series of (i) the original traffic original traffic, and (ii) the predicted traffic predicted traffic in the week from December 16 to 22;

1. **Multi steps** prediction uses a given trained LSTM model for the area a and the corresponding test data to implements a recursive forecasting method based on the ***algorithm above***, where the output of the previous forecast is then fed into the next, as shown below.
 - (a) $\text{prediction}(t) = \text{model}(\text{obs}(t-n), \dots, \text{obs}(t-2), \text{obs}(t-1))$
 - (b) $\text{prediction}(t+1) = \text{model}(\text{obs}(t-n+1), \dots, \text{obs}(t-1), \hat{x}_a(t))$
 - (c) $\dots$ Until the end of test data

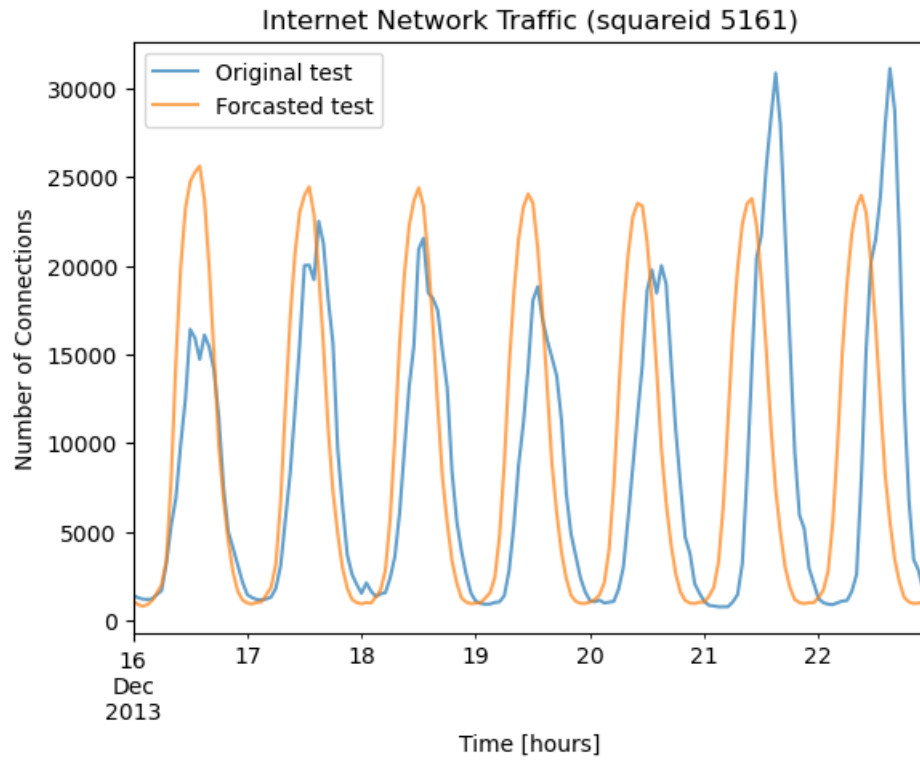


Figure 6: Observed and forecasted internet traffic from December 16 to 22 (Square id 5161)

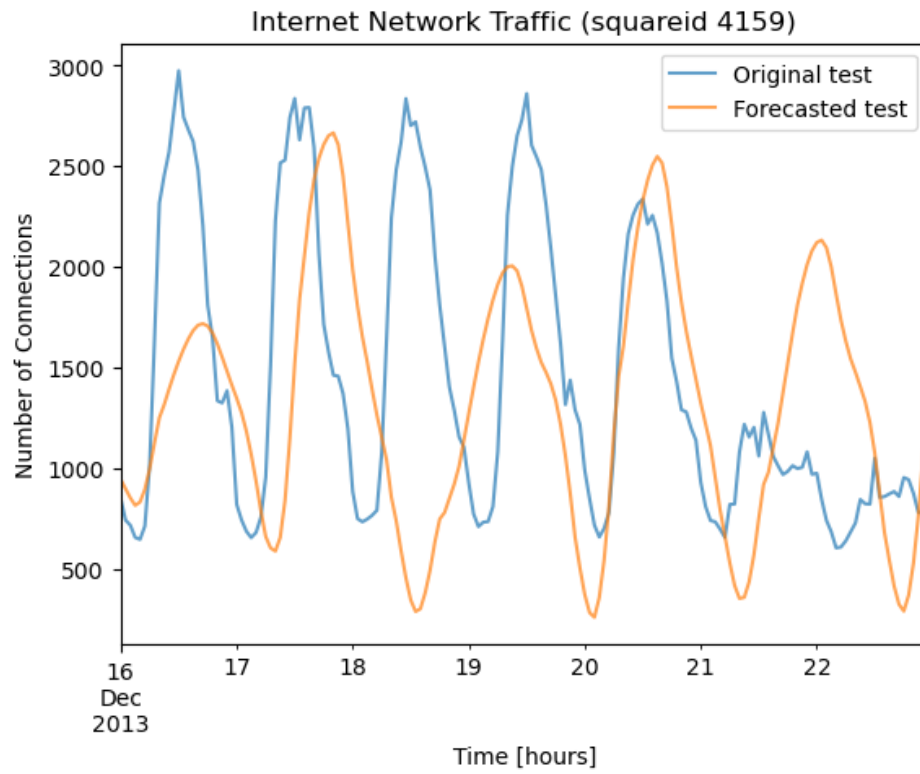


Figure 7: Observed and forecasted internet traffic from December 16 to 22 (Square id 4159)

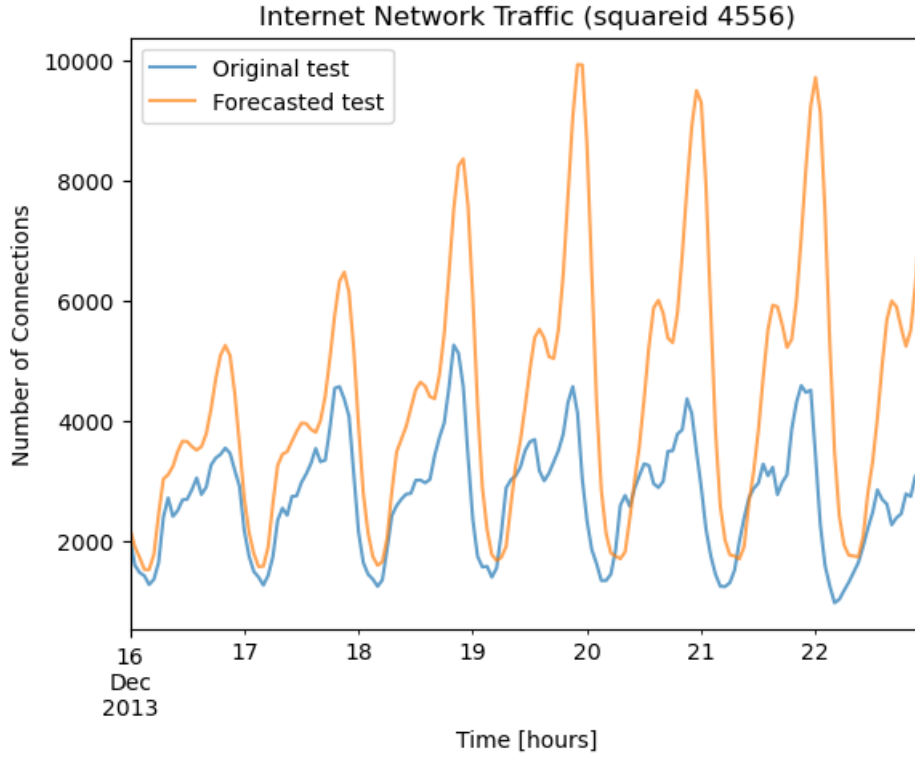


Figure 8: Observed and forecasted internet traffic from December 16 to 22 (Square id 4556)

3.3 Model performance

A table reporting the mean absolute error mean absolute error (MAE) and the mean absolute percentage error mean absolute percentage error (MAPE) computed for the time series at the second item above;

Square id	MAPE(%)	MAE
5161	144.4	5957.83
4159	51.8	680.48
4556	67.6	1769.7

Table 1: Model performance

3.4 Training statistics

Exact statistics on the training and execution time statistics on the training and execution time of the algorithm, with details on the process used to compute such statistics and on the hardware on which they were recorded;

Square id	LSTM training time (seconds)
5161	65.4795548915863
4159	65.72150254249573
4556	64.91609358787537

Table 2: LSTM moel training time for each area

3.5 Personal considerations

A short text where the applicant provides personal considerations on the design, performance, and possible margins for improvement of the algorithm.

1. Since we have not properly addressed seasonality and stationarity issues in the time series before training the LSTM, the model’s performance may depend on the geographical area (e.g., hourly time series include daily and weekly seasonality depending on the region).
2. We should first conduct a model selection phase for time series forecasting. This includes testing classical models such as ARIMA and SARIMA, machine learning models such as XGBoost, and deep learning models (e.g., LSTM).
3. Fine-tune the selected model across several geographical areas and determine the optimal hyperparameters for the final model.
4. Fine-tune the input window size (e.g., 24 hours, 48 hours).
5. For multi-step prediction, train the LSTM separately for each forecast horizon, as recursive methods tend to propagate one-step prediction errors over longer horizons.

References

1. Gianni Barlacchi, Marco De Nadai, Roberto Larcher, Antonio Casella, Cristiana Chitic, Giovanni Torrisi, Fabrizio Antonelli, Alessandro Vespignani, Alex Pentland, and Bruno Lepri, “A multi-source dataset of urban life in the city of Milan and the Province of Trentino.” SciData 2, 150055 (2015). <https://doi.org/10.1038/sdata.2015.55>

2. [https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi: 10.7910/DVN/EG](https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/EG)
3. [https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi: 10.7910/DVN/QJ](https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/QJ)

Resources

Pandas

- Ways to Filter Pandas DataFrame by Column Values
<https://www.geeksforgeeks.org/pandas/ways-to-filter-pandas-dataframe>

Time Series & Machine Learning Websites

- Seasonality Detection in Time Series Data
<https://www.geeksforgeeks.org/machine-learning/seasonality-detection>
- Intro to Time Series Forecasting (Kaggle)
<https://www.kaggle.com/code/iamleonie/intro-to-time-series-forecasting>
- TensorFlow Time Series Tutorial
https://www.tensorflow.org/tutorials/structured_data/time_series
- LSTM for Time Series Forecasting
<https://medium.com/@vineet.pandya/use-tensorflow-lstm-for-time-series>
- Seasonal Time Series Forecasting
<https://www.blog.trainindata.com/seasonal-time-series-forecasting/>
- Reshape Input Data for LSTM (Keras)
<https://machinelearningmastery.com/reshape-input-data-long-short-term-memory/>
- Improving Recursive Time Series Forecasting
<https://medium.com/data-science/how-to-improve-recursive-time-series-forecasting>
- Video: Time Series Forecasting
<https://www.youtube.com/watch?v=xaIA83x5Icg>

Scientific Article

1. E. N. Lazarus and A. A. Du Plessis, “Utility-Scale Solar Power Forecasting: Single-Step vs. Multi-Step Feed Forward Neural Networks,” *2023 IEEE International Conference on Environment and Electrical Engineering and 2023 IEEE Industrial and Commercial Power Systems Europe (EEEIC / I&CPS Europe)*, Madrid, Spain, 2023, pp.

1–6.

DOI: <https://doi.org/10.1109/EEEIC/ICPSEurope57605.2023.10194660>

Open Source Code

- LSTM Framework for Time Series Prediction
<https://towardsdatascience.com/lstm-framework-for-univariate-time-se>
- LSTM Stock Prediction Repository
<https://github.com/Joshwen7947/LSTM-Stock-Prediction->
- Time Series Forecasting LSTM Notebook
<https://github.com/huevosabio/ts-predict/blob/master/TS%20Forecastin%20Single-Step%20LSTM%20RNN.ipynb>

Libraries

Pandas

- Official API Reference
<https://pandas.pydata.org/docs/reference/api/>
- Merging and Joining
https://pandas.pydata.org/docs/user_guide/merging.html
- Add Header Row
<https://www.geeksforgeeks.org/python/how-to-add-header-row-to-a-pand>
- Read CSV Files
<https://www.geeksforgeeks.org/pandas/python-read-csv-using-pandas-re-csv/>
- GroupBy and Sum
<https://www.geeksforgeeks.org/python/pandas-groupby-and-sum/>
- Convert Timestamp to Datetime
<https://stackoverflow.com/questions/9744775/how-to-convert-integer-t>
- RangeIndex to DatetimeIndex
<https://stackoverflow.com/questions/48248239/pandas-how-to-convert-r>
- Move Index to Column
<https://medium.com/@heyamit10/ways-to-use-pandas-move-index-to-colum>
- Video Tutorial
<https://www.youtube.com/watch?v=M4qhlzrCXg>

Seaborn

- Distribution Plots Tutorial
<https://seaborn.pydata.org/tutorial/distributions.html>
- KDE Plot Documentation
<https://seaborn.pydata.org/generated/seaborn.kdeplot.html>
- Colormaps in Heatmaps
<https://www.geeksforgeeks.org/python/colormaps-in-seaborn-heatmaps/>

General Web Sources

- W3Schools Python Tutorial
<https://www.w3schools.com/python>
- Load Text File into Pandas
<https://medium.com/data-science/txt-to-pandas-df3aeaf92548>
- Filter Pandas DataFrame by Column Value
<https://medium.com/@agusabdulrahman/how-to-filter-a-pandas-dataframe>
- Unix Hardware and OS Info
<https://www.isye.gatech.edu/kb/determining-hardware-and-os-info-unix>